

Mixcyclopedia



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Introduction:

Overview:

Mixcyclopedia is an application that allows users to explore the world of mixology and cocktail creations. With a large collection of various cocktail recipes, users are able to find the perfect drink to enjoy or make for a crowd.

The main function of Mixcyclopedia is to show the user which cocktails they are able to make given the ingredients they currently have available to them. The application provides menus where the user can select which ingredients they own. After selecting ingredients, the user will be shown all of the cocktail recipes stored in the application which the user can create.

Although seeing all cocktails that are possible to make with a set of ingredients is useful and allows the users to create new and unique cocktails without leaving the comfort of their home, some users may want to search for cocktails with a subset of ingredients to hone in on a specific flavor. Another major function of mixcyclopedia is getting ideas for cocktails, which may contain some ingredients that the user does not own. This function could be useful if the user is looking for new cocktails that may need additional ingredients which they do not own, or for users who want to find cocktails with a specific flavor, such as lemon juice.

Along with a large amount of prepackaged recipes for the user to explore, Mixcyclopedia allows users to create their own recipes, modify existing recipes, or even delete recipes from the application. Users will be able to define new recipes and add them to the application's recipe collection, or overwrite recipes within the application to contain ingredient proportions that suit the user's preferences. Deleting recipes will also be possible, if the user would like to remove recipes they do not enjoy, or remove a recipe that was mistakenly added.

Objectives:

The objectives of Mixcyclopedia are to:

- Allow the user to view a collection of cocktail recipes that they can make given a set of ingredients they own

- Allow the user to search for cocktail ideas by first selecting which ingredients they want the cocktail to contain, then showing them a collection of cocktails that contain any number of the selected ingredients. This collection of recipes differs from the previous objective because the recipes may contain ingredients that were not selected

- Allows the user to browse the entire collection of recipes locally stored by the application

- Allows the user to add their own recipes, or overwrite existing recipes with their preferred proportions or ingredients

Significance:

Mixcyclopedia provides an easy and convenient way to quickly find cocktails to make with available ingredients.

Whether looking for a new recipe to try out or finding a drink to impress guests, mixcyclopedia provides a fast and easy solution.

Intended Audience:

Any individual above the legal drinking age who would like to explore the world of cocktails and find new and interesting recipes. Specifically, individuals who own a variety of spirits and alcohols, but do not know many cocktail recipes, would gain the most value from the application.

Requirements Analysis:

Necessary Libraries

Built In Libraries

- tkinter : for GUI

Third Party Libraries

- PIL : for using and displaying images in the application

-pysonDB : for storing and retrieving local data

System Requirements

Main Page

1.0.0 User should see frame with application instructions

1.1.0 User should be able to click “show possible” button

1.1.1 User should get redirected to show recipes page

1.1.2 A collection of compatible recipes will be sent to display

1.2.0 User should be able to click “show ideas” button

1.2.1 User should get redirected to show recipes page

1.2.2 A collection of compatible recipes will be sent to display

1.3.0 User should be able to click “show all” button

1.3.1 User should get redirected to show recipes page

1.3.2 A collection of all recipes will be sent to display

1.4.0 User should be able to click “add recipe” button

1.4.1 User should get redirected to add recipe page

1.5.0 A section of selectable ingredients should be displayed for the user

1.5.1 A panel of selectable liquor ingredients will be shown with all locally stored liquor ingredients

1.5.2 A panel of selectable other alcohol ingredients will be shown with all locally stored other alcohol ingredients

1.5.3 A panel of selectable mixer ingredients will be shown with all locally stored mixer ingredients

Show Recipes

2.0.0 A panel of recipes should be displayed for the user

2.0.1 if passed in recipes list is empty display message that no recipes match with selected ingredients

2.0.2 if recipes exist in passed in recipes collection, display all recipes with recipe display objects

2.0.2.1 recipe display object should display recipe name

2.0.2.2 recipe display object should display list of recipe ingredients with ingredient amount and unit next to each ingredient

2.0.2.3 recipe display object should display a list of recipe garnishes

2.0.2.4 recipe display object should display an image for recipe glass type

2.0.2.5 recipe display object should display a delete button which will delete recipe from screen and db

2.1.0 user should be able to click return button

2.1.1 user should be redirected to main page

Add Recipes

3.0.0 user should be shown add recipe form

3.1.0 user should be able to enter a recipe name

3.1.1 a string entry should be shown to enter recipe name

3.2.0 user should be able to enter a variable number of ingredients

3.2.1 user should be able to click add ingredient button

3.2.1.1 a new ingredient input object is added

3.2.2 user should be able to fill ingredient input object

displayed 3.2.2.0 string entry for ingredient name should be

3.2.2.1 drop down menu entry for ingredient type should be displayed

3.2.2.2 string entry for ingredient amount should be displayed

3.2.2.3 string entry for ingredient unit should be displayed

3.2.2.4 a delete button should be displayed which removes the ingredient input

3.3.0 user should be able to enter a variable number of optional garnishes

3.4.0 user should be able to select recommend glass type from a list of glass types

3.5.0 input verification should be performed to ensure entered recipe is proper

3.5.1 exception is raised when recipe name is blank

3.5.2 exception is raised when no ingredients are added

3.5.3 exception is raised when ingredient input is invalid

blank 3.5.3.0 exception is raised when ingredient name is blank

3.5.3.1 exception is raised when ingredient amount is

3.5.3.2 exception is raised when ingredient type is not selected

3.6.0 user should be able to click save recipe button

3.6.1 entered recipe should be verified then saved to storage

3.6.2 a popup alerts the user that the recipe has been saved successfully

3.7.0 user should be able to click return button

3.7.1 user should be redirected to main page

Db in

4.0.0 Should read all recipes from file and return them as a list

4.1.0 Should read all ingredients from file and return them as a list

4.1.1 should read all liquor ingredients from file and return them as a list

4.1.2 should read all other alcohol ingredients from file and return them as a list

4.1.3 should read all mixer ingredients from file and return them as a list

4.2.0 Should read all previously selected ingredients from file and return them as a list

Db out

5.0.0 Should write a new recipe to file

5.1.0 Should write new ingredients to file

5.1.1 should not write existing ingredient to file

5.2.0 Should write a list of all selected ingredients to file

Application window (mixcyclopedia.py)

6.0.0 A tkinter window is created

6.0.1 Window has proper geometry and is not resizable

6.0.2 Window has proper title

6.0.3 Window has proper logo

6.1.0 A tkinter window is given a background

6.2.0 Large logo is placed at top middle of window

6.3.0 A display frame is defined

6.3.1 display frame can be set to main page

6.3.2 display frame can be set to show recipes page

6.3.3 display frame can be set to add recipe page

6.4.0 A popup alert can be placed in middle of screen

6.4.1 An error alert can be placed (red border)

6.4.2 A success alert can be placed (green border)

System Architecture:

The application will require a data folder to store all json data, and an images folder for the application images and icon. The source code of the application will be structured in 4 modules.

Project File Tree:

Mixcyclopedia Project Folder

```
| - run.pyw
| - setup.py
|
| - data
|   - - glasses.json
|   - - ingredients.json
|   - - recipes.json
|
| - img
|   - - glasses
|   - - - <all glass png images>
|   - - bg.png
|   - - large_logo.png
|   - - small_logo.ico
|   - - spilled_logo.png
|
| - src
|   - - __init__.py
|   - - mixcyclopedia.py
|
|   - - custom_widgets
|   - - - __init__.py
|   - - - custom_widgets.py
|
|   - - database_io
|   - - - __init__.py
|   - - - db_in.py
|   - - - db_out.py
|
|   - - displays
|   - - - __init__.py
|   - - - mix_main_page.py
```

```
| - - - mix_add_recipe_page.py  
| - - - mix_show_recipe_page.py  
|  
| - - input_verification  
| - - - __init__.py  
| - - - input_verification.py
```

Data Design:

Ingredients:

Ingredients attributes:

The Ingredients pyson database will hold ingredients with four attributes. The first attribute will be the name and will hold the name of the ingredient. The second attribute is type and can either be "liquor", "other alcohol", or "mixer". The third attribute is a boolean with the name prev_selected. It will be assigned true if the user previously selected the ingredient. Each ingredient object in the db is also assigned an id number.

Example pyson db file:

```
{
  "data": [
    {
      "name": "Tequila",
      "type": "liquor",
      "prev_selected": false,
      "id": 158451677536709652
    },
    {
      "name": "Lime Juice",
      "type": "mixer",
      "prev_selected": true,
      "id": 289923565785980370
    },
    {
      "name": "Orange Liqueur",
      "type": "other alcohol",
      "prev_selected": false,
      "id": 217102371867280458
    },
  ],
}
```

Drink Recipes:

Drink recipe attributes:

- name: name of cocktail recipe
- ingredients: all of the ingredients to make the cocktail recipe. Each tuple in the ingredients list will contain four values, the ingredient name, how much of the ingredient to add, the unit of the ingredient(oz, ml, dash), and the type of the ingredient(liquor, other alcohol, or mixer)
- garnishes: a list of optional garnishes that can be paired with the cocktail to improve appearance
- glass: the name of the recommended glass type for the recipe
- id: id number

Example pyson db file:

```
{
  "data": [
    {
      "name": "Margarita",
      "ingredients": [
        [
          "Tequila",
          "2",
          "oz",
          "liquor"
        ],
        [
          "Lime Juice",
          "1",
          "oz",
          "mixer"
        ],
        [
          "Orange Liqueur",
          "1",
          "oz",
          "other alcohol"
        ],
        [
          "Simple Syrup",
          ".5",
          "oz",
          "mixer"
        ]
      ],
      "garnishes": [
        "Lime Wedge",
        "Salt Rim",
        "Sugar Rim"
      ],
      "glass": "Margarita",
      "id": 680115996927053059
    },
  ],
}
```

Glasses:

glass attributes:

Each glass has three attributes. The name attribute holds the name of the glass. The file_name attribute holds the file name of the image for the glass. And an id number.

Example pyson db file:

```
{
  "data": [
    {
      "name": "Collins",
      "file_name": "Collins.png",
      "id": 192430666931206004
    },
    {
      "name": "Hurricane",
      "file_name": "Hurricane.png",
      "id": 154085907395041231
    },
    {
      "name": "Rocks",
      "file_name": "Rocks.png",
      "id": 248880551258721281
    },
    {
      "name": "Highball",
      "file_name": "Highball.png",
      "id": 109667880826736447
    },
    {
      "name": "Coupe",
      "file_name": "Coupe.png",
      "id": 292733594316931979
    },
    {
      "name": "Copper Mule",
      "file_name": "Mule.png",
      "id": 293020001147385165
    }
  ]
}
```

User Interface Design:

Main Page:

Mixcyclopedia

MIXCYCLOPEDIA LOGO

A brief but informative paragraph of instructions which explains the application, tells the user how to select ingredients from the menu below, and describes the function of each button in the bottom right panel.....Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur. Quis aute iure reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint obcaecat cupiditat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Liquors	Other Alcohols	Mixers
☐ Rum ☐ Gin ☒ Vodka ☐ Tequila ☒ Whiskey	☐ Melon Liqueur ☒ Apple Vodka ☐ Aperol ☒ Honey Liqueur ☒ Peppermint Liqueur ☒ Jägermeister	☒ Lemon Juice ☒ Cola ☐ Soda Water ☐ Lemon Lime Soda ☐ Club Soda ☒ Ginger Ale ☐ Simple Syrup ☐ Strawberry Puree

Mix Button

Find Ideas Button

view all Recipes Button

Add Recipe Button

Recipe Display Page:

Mixcyclopedia

MIXCYCLOPEDIA LOGO

Return Button

Cocktail Recipes

- 1oz Tiquilla
- 1oz Lime Juice
- 1oz Simple Syrup

Optional Garnish:

- Lime Wheel
- Salt Rim

Delete

- 1.5oz Whiskey
- 1oz Lemon Juice
- .5oz Simple Syrup

Optional Garnish:

- Lemon rind
- Maraschino cherry
- Orange slice
- Sugar rim

Delete

- 1oz White Rum
- 1oz Coconut Cream
- 3oz Pineapple Juice

Optional Garnish:

- Pineapple slice
- Maraschino cherry

Delete

Add Recipe Page:

Mixcyclopedia

MIXCYCLOPEDIA LOGO

Recipe Name:

Ingredients:

Name:
Type:
Amount:
Unit:
Delete

Name:
Type:
Amount:
Unit:
Delete

Add Ingredient

Garnish Options (separate with comma):

Glass Type:

Return Button

Add Recipe

Module/Component Design:

Modules Overview:

custom_widgets: hold all custom built tkinter widgets
custom_widgets.py

database_io: handles database operations
db_in.py
db_out.py

displays: holds each display page
mix_main_page.py
mix_add_recipe_page.py
mix_show_recipe_page.py

input_verification: handles input verification and exceptions
input_verification.py

Scripts:

custom_widgets.py

Holds all custom widget classes that inherit from tkinter.Frame. These widgets are used throughout the displays to display large amounts of information.

db_in.py

Contains functions which deal with database retrieval operations. Functions include retrieving all data or specific data from one of the three databases. Also contains functions to return all possible/idea recipes from recipes db given a list of ingredients.

db_out.py

Contains functions which deal with database modifying operations. Functions include adding new objects to the databases, removing objects from the databases, and updating objects in databases.

mix_main_page.py

Contains a main_page class which inherits from tkinter.Frame. Creates the main page of the program, which includes instructions, ingredient selection panels, and redirect buttons. An object of this class is placed into the main application window when the main page needs to be displayed to the user.

mix_add_recipe_page.py

Contains a add_recipe_page class which inherits from tkinter.Frame. Creates the add recipe page of the program, which includes a form for the user to enter in a new recipe. An object of this class is placed into the main application window when the add recipe page needs to be displayed to the user.

mix_show_recipe_page.py

Contains a show_recipe_page class which inherits from tkinter.Frame. Creates the show recipes page of the program, which displays recipes in a scrollable display frame. An object of this class is placed into the main application window when the show recipes page needs to be displayed to the user.

input_verification.py

Defines the InvalidFieldException class which inherits from Exception. Also defines a verify_recipe function which will check if all input fields are valid for a given recipe from the add recipe page, raising an InvalidFieldException when a field is invalid.

mixcyclopedia.py

Defines the mixcyclopedia class which inherits from tkinter.Tk. This class creates the application window, sets the background, logo, icon and title of the window. Places displays from the displays module in the middle of the application window. Has a method which places a popup alert in the middle of the screen

when the user needs to be alerted of an error or successful operation.

run.pyw

Main script for the program. Creates a mixcyclopedia object which starts the application window.

setup.py

Installation setup for the program. Discovers all modules in the project. Creates the mixcyclopedia.egg-info folder which contains setup information.

Testing Strategy:

To properly test all of the system requirements for the Mixcyclopedia application, I plan to use a table to keep track of requirements, and use black box testing to see if each input reveals the expected output.

All of the test cases represent one of the requirements listed in the [System Requirements](#) section.

Taking the role of the user and acting out all required functionality of the application will give a good idea whether the application will give proper outputs and be stable once it is finished. This will make sure that the user experience of the application is smooth and error free. I plan on doing multiple rounds of each test and properly documenting the results in the table below.

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
1	1.0.0	User should see frame with application instructions	The main page is displayed for the user	Application instructions are displayed for the user in the top section of the window	PASS	PASS
2	1.1.0	User should be able to click show possible button	The main page is displayed for the user	User is able to see and interact with the show possible button on the main page	PASS	PASS
3	1.1.1	User should	User clicks the	User is redirected to the show	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
		get redirected to show recipes page	show possible button	recipes page		
4	1.1.2	A collection of compatible recipes will be sent to display	User clicks the show possible button	A proper set of compatible recipes is sent to be displayed on the show recipes page	PASS	PASS
5	1.2.0	User should be able to click "Show ideas" button	The main page is displayed for the user	User is able to see and interact with the Show Ideas button on the main page	PASS	PASS
6	1.2.1	User should get redirected to show recipes page	User clicks on the show ideas button	User gets redirected to the show recipes page	PASS	PASS
7	1.2.2	A collection of compatible recipes will be sent to display	User clicks on the show ideas button	A proper set of recipes is sent to be displayed on the show recipes page	PASS	PASS
8	1.3.0	User should be able to click "Show all" button	The main page is displayed for the user	User is able to see and interact with the show all button on the main page	PASS	PASS
9	1.3.1	User should get redirected to show recipes page	User clicks on the show all recipes button	User gets redirected to the show recipes page	PASS	PASS
10	1.3.2	A collection of all recipes will be sent to display	User clicks on the show all recipes button	A set of all recipes is sent to be displayed on the show recipes page	PASS	PASS
11	1.4.0	User should be able to click "add recipe" button	The main page is displayed for the user	User is able to see and interact with the add recipe button on the main page	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
12	1.4.1	User should get redirected to add recipe page	User clicks on the add recipe button	User gets redirected to the add recipe page	PASS	PASS
13	1.5.0	A section of selectable ingredients should be displayed for the user	The main page is displayed for the user	A section where the user selects ingredients is shown on the bottom left of the main page	PASS	PASS
14	1.5.1	A panel of selectable liquor ingredients will be shown with all locally stored liquor ingredients	The main page is displayed for the user	A panel of selectable liquor ingredients is displayed for the user which includes all liquor ingredients with a selectable check box	PASS	PASS
15	1.5.2	A panel of selectable other alcohol ingredients will be shown with all locally stored other alcohol ingredients	The main page is displayed for the user	A panel of selectable other alcohol ingredients is displayed for the user which includes all other alcohol ingredients with a selectable check box	PASS	PASS
16	1.5.3	A panel of selectable mixer ingredients will be shown with all locally stored mixer ingredients	The main page is displayed for the user	A panel of selectable mixer ingredients is displayed for the user which includes all mixer ingredients with a selectable check box	PASS	PASS
17	2.0.0	A panel of recipes should be displayed for the user	The show recipes page is displayed for the user	A panel to hold all passed in recipes is created and displayed in the center of the show recipes page	PASS	PASS
18	2.0.1	if passed in recipes list is empty display	An empty list of recipes is passed in when	A message is displayed in the recipes panel which tells that no compatible recipes were found	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
		message that no recipes match with selected ingredients	displaying the show recipes page			
19	2.0.2	if recipes exist in passed in recipes collection, display all recipes with recipe display objects	A non-empty list of recipes is passed in when displaying the show recipes page	All recipes from the passed in recipes list are displayed in the recipes panel	PASS	PASS
20	2.0.2.1	recipe display object should display recipe name	Show recipes page is displayed, with non-empty recipes list	The name of each recipe from the passed in recipes list is displayed	PASS	PASS
21	2.0.2.2	recipe display object should display list of recipe ingredients with ingredient amount and unit next to each ingredient	Show recipes page is displayed, with non-empty recipes list	The list of ingredients for each recipe in the recipe list is displayed	PASS	PASS
22	2.0.2.3	recipe display object should display a list of recipe garnishes	Show recipes page is displayed, with non-empty recipes list	The list of optional garnishes for each recipe in the recipe list is displayed	PASS	PASS
23	2.0.2.4	recipe display object should display an image for recipe glass type	Show recipes page is displayed, with non-empty recipes list	An image of the recommended glass type is displayed for each recipe in the recipe list	PASS	PASS
24	2.0.2.5	recipe display object should display a delete button	User clicks on delete recipe button on recipe display	Recipe widget is removed from screen and recipe is deleted from database	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
		which will delete recipe from screen and db				
25	2.1.0	user should be able to click return button	Show recipes page is displayed for the user	User is able to see and interact with a return button	PASS	PASS
26	2.1.1	user should be redirected to main page	User clicks on return button	User is redirected to the main page of the application	PASS	PASS
27	3.0.0	user should be shown add recipe form	Add recipe page is displayed for the user	The add recipe form is created and displayed for the user	PASS	PASS
28	3.1.0	user should be able to enter a recipe name	Add recipe page is displayed for the user	A section of the form for entering the recipe name should be displayed in the add recipe form	PASS	PASS
29	3.1.1	a string entry should be shown to enter recipe name	Add recipe page is displayed for the user	A text entry box should be displayed for the user to enter the recipe name	PASS	PASS
30	3.2.0	user should be able to enter a variable number of ingredients	Add recipe page is displayed for the user	User is able to see and enter ingredient entry objects	PASS	PASS
31	3.2.1	user should be able to click add ingredient button	Add recipe page is displayed for the user	User should be able to see and interact with an add ingredient button	PASS	PASS
32	3.2.1.1	a new ingredient input object is added	User clicks on add ingredient button	A new ingredient input object is added to the recipe form	PASS	PASS
33	3.2.2	user should be able to fill ingredient input object	Ingredient input object is created and displayed	User is able to see and enter information in the new ingredient input object	PASS	PASS
34	3.2.2.0	string entry for	Ingredient input	User is able to see and enter	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
		ingredient name should be displayed	object is created and displayed	information in a text box for the ingredient name		
35	3.2.2.1	dropdown menu entry for ingredient type should be displayed	Ingredient input object is created and displayed	User is able to see and select an option from the ingredient type dropdown menu	PASS	PASS
36	3.2.2.2	string entry for ingredient amount should be displayed	Ingredient input object is created and displayed	User is able to see and enter information in a text box for the ingredient amount	PASS	PASS
37	3.2.2.3	string entry for ingredient unit should be displayed	Ingredient input object is created and displayed	User is able to see and enter information in a text box for the ingredient unit	PASS	PASS
38	3.2.2.4	a delete button should be displayed which removes the ingredient input	Ingredient input object is created and displayed	A delete button is created and displayed on the right of the ingredient object which will remove the current ingredient when clicked	PASS	PASS
39	3.3.0	user should be able to enter a variable number of optional garnishes	Add recipe page is displayed for the user	A text entry box is displayed for the user to enter optional garnishes for the recipe	PASS	PASS
40	3.4.0	user should be able to select recommend glass type from a list of glass types	Add recipe page is displayed for the user	A list of all glass types is displayed for the user which allows them to select a recommended glass type for the recipe	PASS	PASS
41	3.5.0	input verification should be performed to ensure	User attempts to save recipe	Input verification is performed to make sure that recipe name, ingredients, garnishes, and glass type are entered and valid	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
		entered recipe is proper				
42	3.5.1	exception is raised when recipe name is blank	User clicks save recipe button when no recipe name is entered	Exception is raised; a popup alerts the user of the error	PASS	PASS
43	3.5.2	exception is raised when no ingredients are added	User clicks save recipe button when no ingredients are entered	Exception is raised; a popup alerts the user of the error	PASS	PASS
44	3.5.3	exception is raised when ingredient input is invalid	User clicks save recipe button when at least one ingredient input has an invalid input field	Exception is raised; a popup alerts the user of the error	PASS	PASS
45	3.5.3.0	exception is raised when ingredient name is blank	User clicks save recipe button when an ingredient input has no name entered	Exception is raised; a popup alerts the user of the error	PASS	PASS
46	3.5.3.1	exception is raised when ingredient amount is blank	User clicks save recipe button when an ingredient input has no amount entered	Exception is raised; a popup alerts the user of the error	PASS	PASS
47	3.5.3.2	exception is raised when ingredient type is not selected	User clicks save recipe button when an ingredient input has no type selected	Exception is raised; a popup alerts the user of the error	PASS	PASS
48	3.6.0	user should be able to click save recipe button	Add recipe page is displayed for the user	User is able to see and interact with a save recipe button	PASS	PASS
49	3.6.1	entered recipe	User clicks save	User entered recipe information is	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
		should be verified then saved to storage	recipe button, all inputs are valid	verified then sent to be written to file		
50	3.6.2	a popup alerts the user that the recipe has been saved successfully	User clicks save recipe button, all inputs are valid	a popup alerts the user that the recipe has been saved successfully	PASS	PASS
51	3.7.0	user should be able to click return button	Add recipe page is displayed for the user	User is able to see and interact with a return button	PASS	PASS
52	3.7.1	user should be redirected to main page	User clicks on return button	User is redirected to the main page of the application	PASS	PASS
53	4.0.0	Should read all recipes from file and return them as a list	A call is made to retrieve all recipes from local storage	All recipes from local storage are read and sent to caller as a list	PASS	PASS
54	4.1.0	Should read all ingredients from file and return them as a list	A call is made to retrieve all ingredients from local storage	All ingredients from local storage are read and sent to caller as a list	PASS	PASS
55	4.1.1	should read all liquor ingredients from file and return them as a list	A call is made to retrieve all liquor ingredients from local storage	All liquor ingredients from local storage are read and sent to caller as a list	PASS	PASS
56	4.1.2	should read all other alcohol ingredients from file and return them as a list	A call is made to retrieve all other alcohol ingredients from local storage	All other alcohol ingredients from local storage are read and sent to caller as a list	PASS	PASS
57	4.1.3	should read all mixer ingredients	A call is made to retrieve all mixer ingredients from	All mixer ingredients from local storage are read and sent to caller as a list	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
		from file and return them as a list	local storage			
58	4.2.0	Should read all previously selected ingredients from file and return them as a list	A call is made to retrieve all previously selected ingredients from local storage	All previously selected ingredients from local storage are read and sent to caller as a list	PASS	PASS
59	5.0.0	Should write a new recipe to file	A call is made to write a recipe to local storage	The new recipe information is properly formatted and written to file	PASS	PASS
60	5.1.0	Should write new ingredients to file	A call is made to write a new ingredient to local storage	The new ingredient information is properly formatted and written to file	PASS	PASS
61	5.1.1	should not write existing ingredient to file	A call is made to write an existing ingredient to local storage	Existing ingredient is detected, and not re-written to file; nothing happens	PASS	PASS
62	5.2.0	Should write a list of all selected ingredients to file	A call is made to write all user selected ingredients to local storage	The list of user selected ingredients are formatted and written to file	PASS	PASS
63	6.0.0	A tkinter window is created	User runs application	Application window is created and displayed	PASS	PASS
64	6.0.1	Window has proper geometry and is not resizeable	User runs application	Window is 1100pxx650px and cannot be resized	PASS	PASS
65	6.0.2	Window has proper title	User runs application	Window has the title "Mixcyclopedia"	PASS	PASS
66	6.0.3	Window has proper logo	User runs application	Window has the Mixcyclopedia logo as icon	PASS	PASS

Test Case	Requirement Tested	Rationale	Input	Expected Output	Round 1	Round 2
67	6.1.0	A tkinter window is given a background	User runs application	Window has a black and white background image with a variety of cocktail images	PASS	PASS
68	6.2.0	Large logo is placed at top middle of window	User runs application	Window has a large Mixcyclopedia logo located at top middle	PASS	PASS
69	6.3.0	A display frame is defined	User runs application	Display frame instance variable is defined	PASS	PASS
70	6.3.1	display frame can be set to main page	Transition method is called to show main page	Main page is shown in display frame	PASS	PASS
71	6.3.2	display frame can be set to show recipes page	Transition method is called to show show recipes page	Show recipes page is shown in display frame	PASS	PASS
72	6.3.3	display frame can be set to add recipe page	Transition method is called to show add recipe page	Add recipe page is shown in display frame	PASS	PASS
73	6.4.0	A popup alert can be placed in middle of screen	Popup method is called to create a popup frame	A popup frame is displayed for the user with a given message and dismiss button	PASS	PASS
74	6.4.1	An error alert can be placed (red border)	Popup method is called to create a popup frame with a red border	popup has a red frame	PASS	PASS
75	6.4.2	A success alert can be placed (green border)	Popup method is called to create a popup frame with a green border	popup has a green frame	PASS	PASS

Implementation Plan:

Step	Status	Start Date	Complete date	Expected Complete Date / Deadline
Come up with project Idea	complete ▾	06-03-2024	06-05-2024	06-09-2024
...Come up with application name	complete ▾	06-05-2024	06-05-2024	06-28-2024
...Come up with application objectives	complete ▾	06-05-2024	06-22-2024	06-28-2024
...make application logo and icon	complete ▾	06-05-24	06-10-24	07-24-2024
Find which python GUI library to use	complete ▾	06-05-2024	06-05-2024	06-09-2024
...start learning Tkinter	complete ▾	06-05-2024	07-01-2024	07-24-2024
...create rapid prototypes of ui elements to determine application layout	complete ▾	06-05-2024	07-01-2024	07-24-2024
Complete Documentation	complete ▾	06-05-2024	07-23-2024	07-24-2024
Complete Draft of Documentation for Midterm	complete ▾	06-05-2024	06-25-2024	06-28-2024
...Introduction	complete ▾	06-05-2024	06-22-2024	06-28-2024
...Requirements Analysis	complete ▾	06-05-2024	Draft: 06-30-2024 Final: 07-23-2024	07-24-2024
...System Architecture	complete ▾	06-05-2024	Draft: 06-22-2024 Final: 07-23-2024	07-24-2024
...Data Design	complete ▾	06-05-2024	Draft: 06-22-2024	07-24-2024

Step	Status	Start Date	Complete date	Expected Complete Date / Deadline
			Final: 07-23-2024	
.....Determine what data needs to be stored and retrieved	complete ▾	06-05-2024	Draft: 06-22-2024 Final: 07-02-2024	07-24-2024
...User Interface Design	complete ▾	06-09-2024	06-20-2024	06-28-2024
.....Draw Wireframes	complete ▾	06-09-2024	06-20-2024	06-28-2024
...Module/Component Design	complete ▾	06-05-2024	Draft: 06-22-2024 Final: 07-23-2024	07-24-2024
...Testing Strategy	complete ▾	06-23-2024	Draft: 06-25-2024 Final: 07-23-2024	07-24-2024
...Implementation Plan	complete ▾	06-19-2024	06-25-2024	06-28-2024
Code all scripts	complete ▾	06-10-2024	07-21-2024	07-24-2024
...custom_widget mod	complete ▾	06-18-2024	07-20-2024	07-24-2024
.....custom_widget.py	complete ▾	06-17-2024	07-20-2024	07-24-2024
...database_io mod	complete ▾	06-17-2024	07-04-2024	07-24-2024
.....db_in.py	complete ▾	06-10-2024	07-04-2024	07-24-2024
.....db_out.py	complete ▾	06-28-2024	07-04-2024	07-24-2024
...displays mod	complete ▾	06-17-2024	07-20-2024	07-24-2024
.....mix_main_page.py	complete ▾	06-17-2024	07-20-2024	07-24-2024
.....mix_add_recipe_page.py	complete ▾	06-17-2024	07-20-2024	07-24-2024

Step	Status	Start Date	Complete date	Expected Complete Date / Deadline
.....mix_show_recipe_page.py	complete ▾	06-18-2024	07-20-2024	07-24-2024
...input_verification mod	complete ▾	06-17-2024	07-12-2024	07-24-2024
.....input_verification.py	complete ▾	06-17-2024	07-12-2024	07-24-2024
...mixcyclopedia.py	complete ▾	06-30-2024	07-20-2024	07-24-2024
...run.pyw	complete ▾	07-21-2024	07-21-2024	07-24-2024
...setup.py	complete ▾	07-21-2024	07-21-2024	07-24-2024
Test requirements	complete ▾	07-21-2024	07-23-2024	07-24-2024
Finish Project	complete ▾	06-03-2024	07-24-2024	07-24-2024